Joint Response to CMS RFI

Scaling Functional and Lifestyle Medicine through AI-Powered, Interoperable Precision Health Platforms

Submitted by: N2V Life and WaiveDx

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Stephanie Carlton
Deputy Administrator
Centers for Medicare & Medicaid Services (CMS)
Department of Health and Human Services

Subject: Response to CMS Request for Information on Digital Health and Interoperability

Dear Deputy Administrator Carlton,

On behalf of N2V Life and WaiveDx, we respectfully submit our joint response to the CMS Request for Information on strengthening the digital health ecosystem. We present a vision for integrating lifestyle medicine, precision diagnostics, and tokenized incentive models through a federated AI and interoperability-first framework.

Our approach aligns with the 21st Century Cures Act, TEFCA, USCDI standards, and CMS value-based care modernization goals. This submission outlines how CMS can catalyze scalable pilots and reimbursement models that improve chronic disease outcomes and reduce healthcare costs nationwide.

We look forward to engaging with your office to advance a shared mission for equitable, outcomes-driven care.

Sincerely,

Wayne Miller

Wayne Miller

Chief Executive Officer, N2V Life

David Stephanich

David Stephanich

Chief Technology Officer, WaiveDx

Executive Summary

N2V Life and WaiveDx propose a scalable precision health infrastructure integrating AI agents, genomics, linked registries, and tokenized incentives to modernize chronic disease prevention. Anchored in lifestyle medicine and data interoperability, our approach allows CMS to align payment with patient improvement, especially in underserved settings.

Key Elements:

- **Diagnostics**: Insert biometrics, Activity, etc Phenotyping (GARS) for addictions
- **Platform**: AI-based care navigation, health coaching, adherence nudging
- **Registry**: Federated technology systems integrated with Cerner/Epic
- **Token Economy**: Token incentives tied to biometric improvement
- **Interoperability**: FHIR, TEFCA, CDS Hooks, USCDI compliance

Advancing Scalable Lifestyle Medicine: A Strategic Imperative for CMS

A. Clinical and Economic Rationale

Lifestyle Medicine (LM) addresses chronic illness through behavioral root causes: diet, sleep, stress, movement, substance use, and social connectedness.

Evidence-Based Outcomes:

- *Type 2 Diabetes*: WFPB diets reverse disease (Barnard, 2009)
- *Cardiovascular Disease*: Ornish's LM program reversed atherosclerosis (Ornish, 1998)
- *Mental Health*: Structured LM reduces medication dependence (Baban & Morton, 2022)

Economic Rationale:

- 86% of U.S. health costs stem from chronic conditions
- American College of Lifestyle Medicine notes that a self-insured plan covering patients with hypertension, hyperlipidemia, diabetes, or combinations thereof, achieved a $9.64 ROI over five years through lowered pharmaceutical use, procedures, and referrals
- Ideal for Medicare Advantage, Medicaid Waivers, and CHART models

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B. Policy Recommendations for CMS

- Create LM CPT bundles for interdisciplinary delivery
- Fund Innovation Center pilots integrating LM in rural clinics

- Track lifestyle adherence in quality metrics (e.g., PROMIS, sleep, HRV)
- Reimburse AI-based LM coaching platforms integrated with EMRs

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C. Integration with N2V Life and WaiveDx

Layer	Function	Details
Diagnostics	Precision Risk Stratification	PGx, biometrics, PROMIS
Platform	AI-Enabled Interventions	Personalized LM nudging, coaching
Registry	Data Aggregation	FHIR-enabled WaiveDx + EHR sync
Incentives	Outcome-Based Rewards	Biometric tokenization
Interoperability	Seamless Data Exchange	Bulk FHIR, TEFCA, CDS Hooks

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D. Implementation Framework

Phase 1: Pilot Sites (Years 1–2)
Urban + rural clinics using N2V-WaiveDx stack integration

Phase 2: AI Learning (Years 2–3)
Federated models refine care matching and payment prediction

Phase 3: National Scaling (Years 3–5)
Deploy to FQHCs, MA plans, and CMS-aligned ACOs

E. Sample Outcome Targets (12 Months)

Metric	Baseline	Target	Source
HbA1c	8.1%	<6.5%	EHR
Medication Spend	$9,200	<$5,000	CMS claims
Hospitalization Rate	2.3/year	<1.0/year	Registry
PROMIS Wellbeing Score	45	>60	Patient-reported
LM Adherence	47%	>80%	AI-tracked

PATIENTS AND CAREGIVERS- RESPONSES

1. What policies or innovations could encourage individuals and caregivers to use digital health tools?

- Deploy AI health agents offering lifestyle medicine coaching, real-time biometric feedback (e.g., HRV, glucose, sleep), and education embedded in mobile platforms.
- Use tokenized rewards (e.g., $HRX token) to incentivize measurable health improvements via PROMIS and biometric data.
- Provide mobile-first, FHIR-based access to medical records via Blue Button 2.0.

2. How can digital health tools better support health equity and reduce disparities in care?

- Equip rural and underserved populations with low-bandwidth, device-agnostic platforms.
- Utilize digital identity and consent frameworks to empower marginalized groups to own and monetize their data.
- Deploy AI coaching agents trained to recognize social determinants of health (SDOH) and route patients to local services.

3. What are the barriers to using these tools?

- Digital literacy, lack of personalization, and platform fragmentation.
- Our system addresses this with adaptive UI, predictive personalization via federated learning, and centralized dashboards integrated across FHIR and TEFCA frameworks.

PROVIDERS-RESPONSES

1. What are the barriers to implementing digital health tools?

- Lack of EHR integration and provider bandwidth.
- Our solution embeds LM protocols and decision support via CDS Hooks into Cerner and Epic, reducing administrative burden and improving workflow efficiency.

2. What policies or innovations could support providers in adopting and using digital health tools?

- Reimbursement for LM-based AI decision aids.
- Inclusion of LM protocol delivery in quality metrics and MIPS incentives.
- Automated outcome documentation from REDCap via ICD/SNOMED/FHIR mappings.

3. How can digital tools assist providers with burden reduction and burnout prevention?

- Automate care plan generation using biometric + genomic data.
- Reduce data re-entry with FHIR-connected REDCap registries.
- Use AI to monitor lifestyle adherence and prompt necessary interventions.

PAYERS- RESPONSES

1. What are barriers to reimbursing for digital health technologies?

- Limited frameworks for preventive outcome valuation.
- Our solution uses tokenized incentive systems tied to verified biometric improvement (e.g., HbA1c, HRV), allowing for cost-effective, VBC-aligned reimbursement models.

2. What payment innovations or demonstrations could improve adoption of digital tools?

- Health-to-Earn™ frameworks where digital adherence and lifestyle outcomes generate tangible health currency, which can be converted into plan-based incentives.
- Use AI-driven forecasting tools to pre-authorize reimbursement based on predicted improvements in high-cost patients.

3. How can CMS use digital tools to expand access to care and improve quality?

- Support expansion of federated AI platforms into rural, tribal, and Medicaid-supported clinics.
- Incentivize lifestyle adherence and data-sharing via tokenized health rewards under CMS pilot frameworks.

TECHNOLOGY DEVELOPERS AND DATA NETWORKS- RESPONSES

1. What standards or policies could enhance interoperability and innovation?

- Mandate FHIR, USCDI v4, CDS Hooks, and NIST 800-63 compliance across all digital vendors.
- Support decentralized architectures and API-first development.

2. How can CMS support access to actionable data and digital identity infrastructure?

- Build on Blue Button 2.0 by allowing patients to authorize federated data aggregation for care optimization.
- Incentivize platforms that utilize token-based patient permissioning and audit trails.

3. How can TEFCA evolve to support health equity and innovation?

- Embed AI-assisted consent tools and LM-focused federated registries within TEFCA exchange frameworks.
- Expand TEFCA to include outcome-based reward systems for patient improvement tracking.

VALUE-BASED CARE ORGANIZATIONS AND PAYMENT MODEL PARTICIPANTS- RESPONSES

1. How can digital tools be embedded in VBC models?

- Link tokenized rewards to shared savings models (e.g., ACO REACH).
- Measure and reimburse on lifestyle adherence metrics and PROMIS outcomes.

2. What data are needed to support effective value-based models?

- Longitudinal biometric data (HRV, HbA1c), lifestyle engagement data, PROMIS scores, and genomics.
- Our federated registry and AI agents consolidate this data for CMS-compatible submission.

3. How can CMS foster broader adoption of digital health in APMs and MA?

- Recognize LM adherence as a performance metric.
- Fund pilot programs that use AI + tokenization to scale outcomes-based payment.
- Create LM innovation tracks within CMMI using tools like those offered by N2V and WaiveDx.

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4. RFI Response Strategy (Summarized)

- **Patients/Caregivers**: Promote biometric-based incentives and access to AI health agents
- **Providers**: Enable automated care planning and EHR-linked LM interventions
- **Payers**: Integrate real-time health data for value-based adjustments
- **Tech Vendors**: Support decentralized architectures and FHIR-compliant pipelines
- **Value-Based Entities**: Link tokenized incentives to CMS-shared savings models

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5. Technical Architecture Overview

A. System Workflow

1. **Patient Data Capture**: HRV, glucose, steps, PROMIS → via wearable or app
2. **Data Stream**: AWS Kinesis → S3 & HealthLake
3. **Analytics**: SageMaker ML → Risk Prediction & LM Guidance
4. **Registries**: WaiveDx↔ FHIR → EMR (Cerner/Epic)
5. **Incentives**: Health improvement → Token ($H) → Wallet
6. **CMS Reporting**: Outcome exports via Blue Button & USCDI standards
7. System complies with HIPAA, TEFCA, and supports low-bandwidth rural deployment.

6. References

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7. Appendix (Available Upon Request)

Appendix A – Tokenized Health Incentive Flow
Details how biometric improvements (HRV, HbA1c) trigger token issuance.

Appendix B – Visual Architecture Overview
Describes AI, registry, and FHIR flows with sample pilot configurations.

Appendix C – Platform Use Cases

- AI-led lifestyle medicine plans
- Genomic diagnostics with coaching
- Predictive token issuance and LM ROI models

Appendix D – Pilot Implementation Summary

- Indiana Regional Medical Center (IRMC) example
- WavieDx ↔ Cerner bridge
- Tribal and rural FQHC expansion readiness